BACKUP ARCHITECTURE

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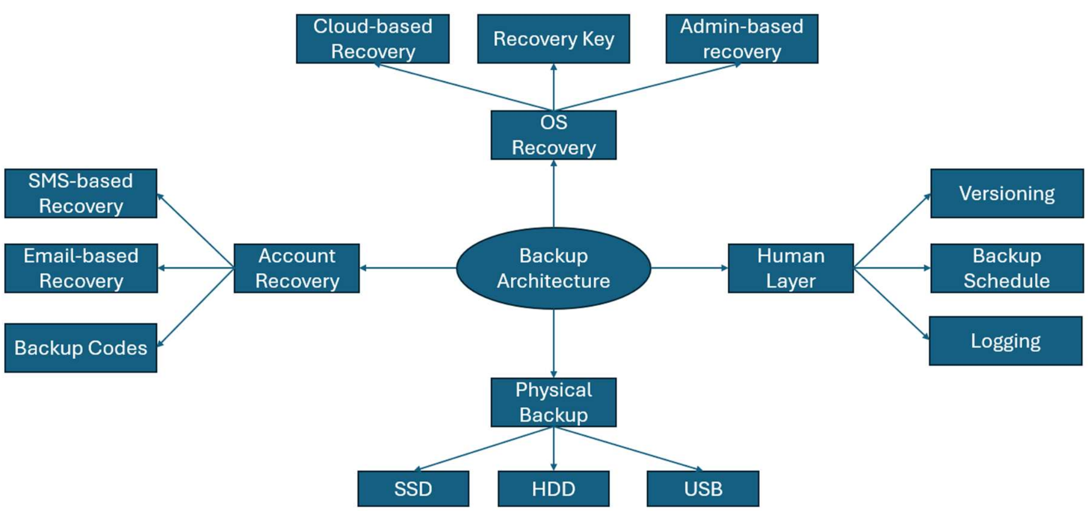
Introduction

This backup architecture is designed to help users build resilient and multi-layered systems for preventing data loss. This is also designed to make backup & recovery knowledge more accessible and easier to understand. This backup architecture was developed as currently data loss is one of the biggest risks and threats to most computer systems currently. This backup architecture is built on the principle that failure can always occur no matter how strong or new a computer system is. This means that computer systems should be made to be resilient and able to recover in the case of failure. This backup architecture covers Cloud, OS, Physical drive recovery and backup methods. Additionally, it also covers automation & human layer along with an example and trade-offs that this backup architecture has.

This backup architecture cannot promise perfect resilience or redundancy within a computer system. It also cannot promise to also prevent data loss within a system as this focuses on recovery mainly. However, this backup architecture can promise to improve resilience and add additional layers of redundancy that reduce the risk of data loss significantly. It is important to remember that each implementation of this architecture will be different depending on purpose and use case. Make sure to adapt the resources and principles of this architecture to your needs.



The spider diagram above shows a visual layout of the domains, recovery, and backup methods mentioned and used in this backup architecture. This diagram does not show all features and methods in the backup architecture. More details can be found in the sections below.

Account and Cloud Recovery & Backup Domain

- **There are 4 main recovery methods that providers offer for recovery of your account.**
- **The first recovery method is recovery phrases**
- **Recovery phrases can be used to reset the account password allowing the user to regain access to their account.**
- **Recovery phrases are user-controlled and cannot be used for MFA as they for password resets only in most cases.**
- **Recovery phrases should be stored encrypted and in multiple locations to prevent an attacker from compromising them or data loss.**
- **I would recommend using recovery phrases if you value control and don’t want a single point of failure using email or SMS.**
- **The second recovery method is backup recovery codes**
- **Backup recovery codes are about 10 one-time usage codes that can be used for resetting MFA on the account.**
- **Backup recovery codes are usually used for authentication app MFA however some websites allow recovery codes to be used for other MFA options.**
- **These codes should be stored encrypted and in multiple locations. I would recommend using backup recovery codes if you use authentication app MFA on your account.**
- **The third recovery method is email-based recovery**
- **Email-based recovery is where you receive a password reset email which allows you to reset your password and regain access.**
- **However, Email-based recovery creates a single point of failure if the same master email is used for recovery on multiple accounts.**
- **I would recommend using email-based recovery on websites that may not offer backup recovery codes or recovery phrases.**
- **The fourth recovery method is SMS-based recovery**
- **SMS-based recovery is where a one-time password reset is sent to your phone number via SMS which can then be used to regain access.**
- **However, SMS-based recovery is vulnerable because of SMS swapping attacks and SMS interception which are commonly used by attackers.**
- **I would recommend only using SMS-based recovery if that is the only option that the service offers although it is unlikely nowadays.**
- **For backups, there are a few cloud providers out there you can use.**
- **For storage, you can use Google Drive, iCloud and OneDrive as their free and paid plans can offer a large amount of storage.**
- **However, these cloud providers can have privacy concerns and shouldn’t be trusted with sensitive data.**
- **For privacy and security, you can use Proton Drive, Sync.com and Tresorit due to their zero-knowledge and end-to-end encryption features.**

OS Recovery and Backup Domain

- **For PC operating systems, there are usually 5 main recovery methods used.**
- **The first recovery method is recovery key.**
- **Recovery keys are used for accessing device data that has been encrypted using device encrypted.**
- **Recovery keys cannot be used for FDE as it uses pre-boot authentication and requires a recovery disk instead.**
- **I would recommend recovery keys if you use device encryption on MacOS, Linux and Windows.**
- **The second recovery method is recovery mode.**
- **Recovery mode is used for software issues in boot configuration and operating processes that require user-input to fix.**
- **Recovery mode depending on the operating system can offer options such as terminal commands, scanning for errors and resetting the device completely.**
- **I would recommend using recovery mode if any errors occur in operating system processes especially registry or firmware.**
- **The third recovery method is admin-based recovery**
- **Admin-based recovery is where users from an admin account reset the password of a standard user or other admins.**
- **Admin-based recovery only works if standard/admin separation is present and is supported by the operating system.**
- **I would recommend using admin-based recovery for companies and enterprise environments as it is effective however not the most secure.**
- **The fourth recovery method is cloud-based recovery**
- **Cloud-based recovery is where users use a connected cloud account to reset their PIN or account password on the operating system.**
- **Cloud-based recovery only works if the operating system supports a connection for example like windows supports Microsoft accounts for security and recovery.**
- **I would recommend cloud-based recovery for quick recovery however this can cause some privacy concerns.**
- **The fifth recovery method is security questions**
- **Security questions are where users make answers to questions like what your mother's maiden name or the name of the first pet is.**
- **These security questions are only supported on some operating systems and are considered the weakest recovery method.**
- **I would only recommend security questions if you have to use it as an option.**
- **For backup, users may use system snapshots, export group policy and other important key data from windows.**
- **It is important to research backup methods that your operating system supports as windows may have more options compared to MacOS.**

Physical Drive Recovery & Backup Domain

- **There are two recovery methods that can be used to recover data using physical drives.**
- **The first recovery method is clean installation via USB**
- **This is where users reinstall the operating system using an USB with a clean copy installed.**
- **This is commonly done when malware is persistently in the operating system or when there are major issues with the current copy of the operating system.**
- **However, USBs cost money and require you to keep a clean copy of the operating system safe from malware.**
- **I would recommend this if you want more redundancy especially with recovery of the operating system.**
- **The second recovery method is data recovery via HDD or SSD**
- **This is where users connect an SSD or HDD with their data to their laptop after it has been reset and transfer the data from the connected drive to the internal drive of the device.**
- **This is commonly done when users completely wipe their PC and need to restore their data without using cloud storage.**
- **However, physical drives cost money especially SSDs which can be expensive and need to be air-gapped to protect against malware infection.**
- **I would recommend this if you want easy data recovery after resetting the PC or reinstalling the operating system.**
- **There are 2 types of physical storage media I will discuss in this domain**
- **The first type is USB**
- **USB otherwise known as universal serial bus is used to transmit data and power between devices.**
- **USBs are commonly used to store clean copies of operating systems and recovery disks by the industry.**
- **You should make sure to encrypt the USB to prevent an attacker from using a recovery-based attack.**
- **I would recommend USB if you use FDE encryption for your device or if you want a nuclear option against malware.**
- **The second type is physical drives**
- **These physical drives are SSD (Solid State Drive) and HDD (Hard Disk Drive) used for data storage both internally and externally.**
- **These are commonly used as device storage with SSD being faster and more secure than HDD.**
- **You should use device encryption or FDE encryption to protect your data.**
- **I would recommend an external physical drive to use an resilient and air-gapped backup for your most sensitive data.**

<u>Automation & Human Domain</u>

- **There are two main types of automation that are commonly used for backups.**
- **The first type of automation is cloud sync.**
- **This is where users download the app version of the cloud storage and have their files uploaded and updated in real-time onto the cloud servers.**
- **Cloud sync is commonly used as it would be too time-consuming and exhausting for people to manually upload their files to cloud storage.**
- **However, this can fill up storage at an uncontrollable rate and be unsecure as there is no application passcode most of the time offered on PC OSs.**
- **I would recommend cloud sync to users or companies that want quick and reliable backups to the cloud.**
- **The second type of automation is custom scripts**
- **This is where users create custom specific scripts to perform backup processes automatically.**
- **Custom scripts are used for OS-level backups as power users may like to handle specific backup processes**
- **However, this can fail silently and requires the IT team or the user to maintain and update these scripts.**
- **I would recommend custom scripts for power users or IT teams in companies that need control over their backup processes.**
- **For the human layer, there are two areas that the human layer.**
- **The first area is manual backups**
- **This is where the user uploads files to cloud storage, performs exports and transfers files to the physical drives.**
- **This is commonly done when the user doesn’t trust automation or isn’t able to write custom scripts that can handle the backup processes they use.**
- **The user can use backup schedules with backups spread across days, a priority system for most urgent backups and logging for tracking what was backed up.**
- **I would recommend manual backups for high-risk users that cannot afford to trust automation that can be compromised or fail silently.**
- **The second area is habits.**
- **This is where the user has good backup hygiene to make sure that their backup system is not only redundant and but resilient.**
- **This is commonly done when the user or IT team needs to ensure that the backups work well and that backups are organised in certain ways.**
- **These habits can include versioning where they have multiple versions of the data and testing backups where they ensure that the backups would work in a real-life situation.**
- **I would recommend having habits for anyone with a backup system as it allows you to play a role in your backup system.**
- **It also makes sure that your backup system is verified and organised clearly.**

Contingency Planning

- **Contingency Planning is where failure is directly planned for within an system or for a situation.**
- **Contingency planning is important to be prepared for failure within the backup system.**
- **The first step is to calm down.**
- **This can be done by taking a few deep breaths and quickly releasing any anger by hitting something.**
- **This step is important to reduce the emotional noise of the stressful, frustrating or even terrifying situation.**
- **The second step is to identify**
- **This can be done by asking yourself the questions like what is the situation? What can I do?**
- **This step is important as it allows you to use a targeted response instead of a generic response.**
- **The third step is containment**
- **This can be done by immediately containing the failure by disconnecting it from the internet or securing backups in other locations.**
- **This step is important to ensure that the backups in the other systems are safe and that the failure is isolated to that component only.**
- **The fourth step is recovery**
- **This can be done by using physical drive, snapshots, recovery keys, codes, USB and recovery mode depending on the failure.**
- **This step is important as the failure can be recovered from and it also avoids potentially having to create a new account or buy a new device.**
- **If recovery were to fail, there are a few actions you can take.**
- **You should start off by making sure to calm down as a recovery failure can be extremely frustrating and stressful.**
- **After calming down, you can attempt other recovery options as its very unlikely that other options won't work.**
- **If recovery is successful with using other options, you can move onto to the final step of this plan.**
- **If recovery is not successful, you can create a new account, buy a new device or get replacements as needed.**
- **The fifth and final step is documentation.**
- **This can be done through the structure what (what was the failure?) → Why (Why did the failure happen?) → Response (How did you respond?) → Potential improvements (What could be improved for next time?)**
- **This step is very important as it creates a feedback loop allowing you to repair and learn from mistakes or failures that happened.**

Backup System Set-up User Workflow

- **First, the user discovers the Backup Architecture while looking for tips on GitHub.**
- **The user decides to download the document and read it to use for themselves.**
- **Second, the user starts with the cloud domain.**
- **The user looks at their data which is a mixture of games and work-related documents.**
- **They decide to use Proton Drive for their work-related documents and Google Drive for their game data.**
- **After creation, the user now has cloud backups and not a single point of failure which used to be their laptop.**
- **Third, the user moves onto the OS domain.**
- **The user uses windows 11 Pro for their main laptop and decides to export their registry, group policy and password policy.**
- **The user then uploads these documents onto their cloud storage which gets quite full but manages to fit.**
- **After uploading to cloud storage, the user decides to use an USB originally used for gaming to store a clean copy of Windows 11.**
- **They encrypt the USB using BitLocker for security and to prevent recovery-based attacks.**
- **Fourth, the user moves onto the Physical Drive domain.**
- **The user uses an HDD they used to use for gaming storage to store backups of their most sensitive data.**
- **This data includes medical information, bank details and OS-level backups as they are extremely sensitive for the user.**
- **The user then encrypts the HDD via BitLocker using FDE encryption to protect from attacker compromise.**
- **Fifth, the user decides to implement automation and the human layer.**
- **The user downloads Proton drive on their laptop and syncs their documents in real-time to the cloud server.**
- **The user also implements versioning with keeping multiple versions of backups just in case the recent backup were to fail.**
- **The user struggles with manually implementing versioning so they ask AI to create a custom script that creates backups of certain data and sorts them out using dates.**
- **Sixth, the user finishes setting up the backup system by implementing the contingency plan structure.**
- **The user adds specific notes for their cloud storage and own mindset so they can respond effectively and efficiently to failures.**
- **It is important to remember that this is an example it will be different for each user to implement.**

<u>Backup Architecture Trade-Offs</u>

- **This backup architecture has trade-offs just like any other system I have created so far.**
- **It is important to remember that trade-offs may vary depending on your implementation of the architecture**
- **I have identified 5 main trade-offs with my backup architecture.**
- **The first trade-off is cost**
- **This trade-off exists because some recovery tools and methods like physical drives, USB and paid tiers on cloud storage can be expensive.**
- **This trade-off can be managed by using free, built in recovery tools, using older hardware if needed and free tiers of cloud storage.**
- **The second trade-off is cognitive load.**
- **This trade-off exists because maintaining or creating a backup system can be exhausting and stressful.**
- **This means that there is more of a risk of human error and burnout for the user or the IT team.**
- **This trade-off can be managed by automation, incremental and targeted implementation and rotation schedules/rest periods.**
- **The third trade-off is time-consumption.**
- **This trade-off exists because creating cloud accounts, exporting operating system data, formatting and encryption physical media and make contingency plans can take weeks or months even.**
- **This means that some users or IT teams may not have the time to be creating the backup system.**
- **This trade-off can be managed by slow incremental updates and implementing backup features based on needs.**
- **The fourth trade-off is compatibility**
- **This trade-off exists because some recovery tools and methods may not be supported on older hardware or operating systems.**
- **For example, MacOS doesn't support admin-based recovery or HDDs may be slower and less effective compared to SSDs.**
- **This trade-off can be managed by using the best recovery methods where possible, second best where the best isn't achievable and using virtual machines.**
- **The fifth trade-off is storage overhead**
- **This trade-off exists because some data can be extremely large especially OS-level data like registry or group policy.**
- **This means that users or IT teams may struggle to fit in backups of data overall decreasing resilient and redundancy.**
- **This trade-off can be managed by paid tier of cloud storage, using physical storage media, device storage and a priority system.**